2.1 The Field Axioms and the First Examples

Number Systems: From $\mathbb{N}$ *to* $\mathbb{R}$ defined fields (Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1) and built $\mathbb{Q}$, $\mathbb{R}$ and, for each prime p, the field $\mathbb{Z}/p\mathbb{Z}$. *Analytic Preliminaries*, Chapter 1 added $\mathbb{C}$. This chapter looks at fields as a kind of object in their own right. Its first section restates the definition under the name that the linear-algebra volumes use, collects the rules that hold in every field, solves one and two linear equations in any field, and sorts the standard examples from the standard non-examples, naming in each case the clause that fails.

2.1.1 What solving equations needs

Here is a small system of equations, solved the way it is solved at school:

$$\begin{aligned} 2x + 3y &= 7, \\ x - y &= 1. \end{aligned}$$

Subtract twice the second equation from the first: $5y = 5$. Divide by 5: $y = 1$. Then $x = 1 + y = 2$. Check: $2 \cdot 2 + 3 \cdot 1 = 7$ and $2 - 1 = 1$.

List the moves. Equations were combined by sums and differences, one equation was scaled by a number, and one was divided by a number that is **not zero**. Silently, terms were also reordered, regrouped, and brackets were multiplied out. Elimination, the method the linear-algebra volumes run on systems of any size, uses these moves and nothing else.

Now stay inside $\mathbb{Z}$. The equation $2x = 1$ has integer coefficients and no integer solution: $2x$ is even and $1 = 2 \cdot 0 + 1$ is odd, and no integer is both (Number Systems: From \(\nN\) to \(\nR\), Corollary 2.5.3 (a)). The move "divide by 2" leads out of $\mathbb{Z}$. So the dividing line between $\mathbb{Z}$ and $\mathbb{Q}$ is division, and division by every non-zero element is what the definition must ask for.

A field is a number system where you can add, subtract, multiply, and divide by anything non-zero, with the usual rules.

2.1.2 The definition

The definition lists the rules once, as clauses that can be checked and cited. The numbering is the one *Number Systems* used for commutative rings, and the one the linear-algebra volumes cite.

Definition 2.1.1. Field

A **field** is a set F together with two functions, **addition** $+\colon F \times F \to F$, $(a, b) \mapsto a + b$, and **multiplication** $\cdot\colon F \times F \to F$, $(a, b) \mapsto a \cdot b = ab$, satisfying the following clauses.

(F1) $a + b = b + a$ **for all** $a, b \in F$.

(F2) $(a + b) + c = a + (b + c)$ **for all** $a, b, c \in F$.

(F3) **There is** an element $0 \in F$ such that $a + 0 = a$ **for every** $a \in F$.

(F4) **For every** $a \in F$ there is an element $-a \in F$ such that $a + (-a) = 0$.

(F5) $(ab)c = a(bc)$ **for all** $a, b, c \in F$.

(F6) $ab = ba$ **for all** $a, b \in F$.

(F7) **There is** an element $1 \in F$ with $1 \neq 0$ such that $a \cdot 1 = a$ **for every** $a \in F$.
(F8) **For every non-zero** $a \in F$ there is an element $a^{-1} \in F$ such that $aa^{-1} = 1$.
(F9) $a(b + c) = ab + ac$ **for all** $a, b, c \in F$.

We write $a - b := a + (-b)$, and $a/b := ab^{-1}$ for $b \neq 0$.

Read the clauses in three groups.

- **(F1)–(F4)** speak only of addition. The order and the grouping of a sum do not matter, 0 changes nothing when added, and each element can be undone by its negative. Subtraction is built from these.
- **(F5)–(F8)** repeat those four statements for multiplication, with one change: an inverse is demanded only of a **non-zero** element. So division is defined only by non-zero elements, and no clause ever calls for division by 0. Clause (F7) also demands that the two neutral elements differ.
- **(F9)**, the distributive law, is the single bridge between the two operations: it alone mentions both, and multiplying out brackets rests on it, with (F6) for a bracket on the left.

Mind the order of the quantifiers in (F3): there is **one** element 0 that works **for every** a. The weaker reading, "each a has its own z with $a + z = a$", would let z change with a, and it is not what (F3) says. In (F4) the order is the other way round, as it has to be: the negative $-a$ depends on a.

Both operations are functions **into** F: a sum or product of two elements of F is again in F. For a candidate field cut out of a larger number system, that closure is where checking starts. We say "F is a field" when the two operations are understood, and write $(F, +, \cdot)$ when they must be named.

Remark (The same definition in *Number Systems*). Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1 calls a set with two operations a **commutative ring** when (F1)–(F7) and (F9) hold, its (F7) asking only for an element 1 with $a \cdot 1 = a$; it calls a commutative ring a **field** when, in addition, $1 \neq 0$ and (F8) hold. Clause by clause, these are the conditions of Definition 2.1.1: the nine clauses are worded alike, and the only difference is where the condition $1 \neq 0$ is filed, inside (F7) here and as a separate clause there. So F is a field in the sense of Definition 2.1.1 **if and only if** it is a field in the sense of Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1. Every result that *Number Systems* proves for fields, or for commutative rings, therefore applies to every field of this chapter, and conversely. The separate filing there has a purpose: it lets the one-element set $\{0\}$ count as a commutative ring (the zero ring) while keeping it out of the fields.

Quick check. Is $\mathbb{Q}_{>0}$, the set of positive rationals with the usual $+$ and $\cdot$ of $\mathbb{Q}$, a field?

Solution. No. Addition and multiplication do map $\mathbb{Q}_{>0} \times \mathbb{Q}_{>0}$ into $\mathbb{Q}_{>0}$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.2.1, with (P1) of Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.1). But (F3) fails. An element $z \in \mathbb{Q}_{>0}$ with $1 + z = 1$ would satisfy $z = 0 + z = ((-1) + 1) + z = (-1) + (1 + z) = (-1) + 1 = 0$ in the field $\mathbb{Q}$, by (F1), (F3), (F4) and (F2), and $0 \notin \mathbb{Q}_{>0}$. (Once (F3) fails, (F4) cannot even be stated.)

2.1.3 Rules that hold in every field

The definition's notation takes something for granted. The symbols 0, $-a$ and a^{-1} are written as names of single elements, while (F3), (F4), (F7) and (F8) assert only that suitable elements **exist**. Were there two different zeros in some field, the symbols 0, $a-b$ and a/b would be ambiguous. The next theorem removes the worry and collects the rules of school algebra that every field obeys. Most of it was proved in *Number Systems* for every commutative ring; we collect it under one name, citing the proofs there, and add the rules for fractions as part (g).

Theorem 2.1.1. Basic Properties of Fields

Let F be a field and let $a, b, c, d \in F$.

(a) The element 0 in (F3) is unique, and the element 1 in (F7) is unique.

(b) (Cancellation) If $a+b=a+c$, then $b=c$. If $ab=ac$ and $a\neq 0$, then $b=c$.

(c) The negative $-a$ in (F4) is unique, and for $a\neq 0$ the inverse a^{-1} in (F8) is unique.

(d) $0\cdot a=0$.

(e) $(-1)a=-a$.

(f) If $ab=0$, then $a=0$ or $b=0$.

(g) If $b\neq 0$ and $d\neq 0$, then $bd\neq 0$, $(bd)^{-1}=b^{-1}d^{-1}$, $(b^{-1})^{-1}=b$, and

$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}, \qquad \frac{a}{b}\cdot\frac{c}{d}=\frac{ac}{bd}.$$

Idea. Work backwards from each goal to the clause that could produce it; each part then takes a few lines. For uniqueness, take two candidates and prove them equal. No clause says anything about $0\cdot a$, so for (d) we manufacture a place where (F9) can act: $0=0+0$, distribute, cancel. In (f) the hypothesis $a\neq 0$ has exactly one use, which is to unlock (F8). The fraction rules in (g) come from one move: an element is identified by showing that it satisfies the defining equation of the element we want, and uniqueness finishes.

Proof. By the remark after Definition 2.1.1, F is a commutative ring in the sense of Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1, so Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 applies to it.

(a) This is Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a).

(b) Suppose $a+b=a+c$. Put $-a$ in front of each side and regroup with (F2): this gives $((-a)+a)+b=((-a)+a)+c$. By (F1) and (F4), $(-a)+a=a+(-a)=0$, so $0+b=0+c$, and by (F1) and (F3) this says $b=c$.
Now suppose $ab=ac$ and $a\neq 0$. By (F8), a^{-1} exists. Using (F7), (F8), (F6) and (F5),

$$\begin{aligned} b &= b\cdot 1 = b(aa^{-1}) = (ba)a^{-1} = (ab)a^{-1} \\ &= (ac)a^{-1} = (ca)a^{-1} = c(aa^{-1}) = c\cdot 1 = c. \end{aligned}$$

(c) The negative is unique by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a). For $a\neq 0$, clause (F8) says that a is a unit of F in the sense of Definition 1.4.1, and the inverse of a unit is unique (the well-definedness paragraph after Definition 1.4.1).

(d) By Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (b), $a \cdot 0 = 0$, and $0 \cdot a = a \cdot 0$ by (F6).

(e) By Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c) with $a = 1$, $(-1)b = -(1 \cdot b)$ for every $b \in F$; and $1 \cdot b = b \cdot 1 = b$ by (F6) and (F7). Applied to $b = a$, this gives $(-1)a = -a$.

(f) Suppose $ab = 0$. If $a = 0$ there is nothing to prove, so suppose $a \neq 0$. By (F8), a is a unit of F (Definition 1.4.1), and a unit is not a zero divisor (the paragraph after Definition 1.4.1): from $ab = 0$ it follows that $b = 0$. So $a = 0$ or $b = 0$.

(g) Let $b \neq 0$ and $d \neq 0$. If $bd = 0$, then (f) would give $b = 0$ or $d = 0$; so $bd \neq 0$. By (F8), b and d are units of F. A product of units is a unit, with $(bd)^{-1} = b^{-1}d^{-1}$, and the inverse of a unit is a unit, with $(b^{-1})^{-1} = b$ (the closure paragraph after Definition 1.4.1). Here $b^{-1} \neq 0$, since $1 \neq 0$ by (F7) and so 0 is not a unit (the paragraph "Why this definition" after Definition 1.4.1); thus $(b^{-1})^{-1}$ is also the inverse that (F8) supplies, by (c).

For the sum, multiply out with (F9) and reorder with (F5) and (F6):

$$\begin{aligned}(ab^{-1} + cd^{-1})(bd) &= a(b^{-1}b)d + c(d^{-1}d)b \\ &= ad + cb = ad + bc,\end{aligned}$$

by (F8) and (F7) and then (F6). Multiplying both ends by $(bd)^{-1}$, and using $(bd)(bd)^{-1} = 1$ with (F5) and (F7) on the left, gives $ab^{-1}+cd^{-1} = (ad+bc)(bd)^{-1}$, which is the first displayed rule. For the product, (F5) and (F6) give $(ab^{-1})(cd^{-1}) = (ac)(b^{-1}d^{-1}) = (ac)(bd)^{-1}$, by the formula for $(bd)^{-1}$ just proved. This proves the theorem.

■

From now on we use $0 + a = a$ and $1 \cdot a = a$ silently; they follow from (F1), (F3), (F6) and (F7). The larger gain is that any school identity whose proof uses only (F1)–(F9) is true in every field: for instance $(a + b)^2 = a^2 + 2ab + b^2$, where $a^2 := aa$ and $2 := 1 + 1$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (d)), and $(-a)(-b) = ab$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c)). Part (f) is the rule we will lean on most: a product of non-zero elements is non-zero.

Remark. Part (g), with the uniqueness of inverses in (c), is the content of Number Systems: From \(\nN\) to \(\nR\), Lemma 4.1.1 (a) and (b), which *Number Systems* proved for every field when it studied ordered fields. Part (f) and the inverse identities of (g) come from the paragraphs on units after Definition 1.4.1; the two fraction formulas are proved here rather than cited from Number Systems: From \(\nN\) to \(\nR\), Lemma 4.1.1, so that no theorem of this section rests on that chapter; only the examples $\mathbb{R}$ and $\mathbb{C}$ below lean on it.

Quick check. In a field, does $a^2 = 0$ imply $a = 0$?

Solution. Yes. $a^2 = aa = 0$, so by Theorem 2.1.1 (f) with $b = a$, $a = 0$ or $a = 0$.

Warning. Cancellation needs its hypothesis. "If $ab = ac$, then $b = c$" is **false** without $a \neq 0$: in $\mathbb{Q}$, $0 \cdot 1 = 0 \cdot 2$, yet $1 \neq 2$. Every time a factor is canceled, say why it is not 0.

Part (f) has a consequence that school algebra uses all the time without naming it: an element has at most two square roots, and they are negatives of each other.

Corollary 2.1.2. Equal Squares

Let F be a field and $x, y \in F$. Then $x^2 = y^2$ if and only if $x = y$ or $x = -y$. In particular, $x^2 = 1$ if and only if $x = 1$ or $x = -1$.

Idea. Move everything to one side and factor: $x^2 - y^2 = (x - y)(x + y)$, the school identity, which holds in every field. A product is zero only when a factor is, by part (f), and the two factors give the two cases.

Proof. First, $(x - y)(x + y) = x^2 - y^2$. Indeed, by (F9) and (F6), $(x + (-y))(x + y) = xx + xy + (-y)x + (-y)y$, and Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c) turns the last two terms into $-(yx)$ and $-(yy)$. Since $yx = xy$ by (F6), the terms xy and $-(xy)$ add to 0 by (F4), and what is left is $x^2 - y^2$.

($\Rightarrow$) If $x^2 = y^2$, then $x^2 - y^2 = y^2 + (-y^2) = 0$ by (F4), so $(x - y)(x + y) = 0$. By Theorem 2.1.1 (f), $x - y = 0$ or $x + y = 0$. In the first case, adding y to both sides gives $x = y$, by (F2), (F1), (F4) and (F3). In the second, y satisfies the defining equation of $-x$, so $y = -x$ by Theorem 2.1.1 (c), and then $x = -(-x) = -y$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a).

($\Leftarrow$) If $x = y$, there is nothing to prove. If $x = -y$, then $x^2 = (-y)(-y) = yy = y^2$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c).

For the last sentence take $y = 1$, with $1^2 = 1$ by (F7).

■

Remark. The two square roots y and $-y$ can coincide even when $y \neq 0$: in the field with two elements of the next section, $1 + 1 = 0$, so $-1 = 1$. And the corollary needs (f), so it can fail in a commutative ring with zero divisors. In $\mathbb{Z}/8\mathbb{Z}$ we have $[3]^2 = [9] = [1]$, yet $[3] \neq [1]$ and $[3] \neq [-1] = [7]$, since 8 divides neither $3 - 1 = 2$ nor $3 - 7 = -4$. For $\mathbb{C}$, *Analytic Preliminaries*, Chapter 1 proved the same fact directly from real and imaginary parts (Analytic Preliminaries, Proposition 1.4.1 (b)); the corollary gives it in every field at once.

2.1.4 Solving linear equations in a field

The opening slogan promised that a field is where elimination works. Here is that promise kept for one equation in one unknown and for two equations in two unknowns.

Proposition 2.1.3. Linear Equations in a Field

Let F be a field and let $a, b, c, d, e, f \in F$.

(a) If $a \neq 0$, the equation $ax = b$ has **exactly one** solution $x \in F$, namely $x = a^{-1}b$.

(b) If $ad - bc \neq 0$, the system

$$\begin{aligned} ax + by &= e, \\ cx + dy &= f \end{aligned}$$

has **exactly one** solution $(x, y) \in F \times F$, namely

$$x = \frac{ed - bf}{ad - bc}, \qquad y = \frac{af - ce}{ad - bc}.$$

Idea. Uniqueness and existence are separate jobs. For uniqueness, assume a solution and run the school elimination on it: one combination of the two equations kills y, another kills x, and each leaves an equation of the form (a). For existence, substitute the candidate and multiply out. The quantity $ad - bc$ is exactly the coefficient elimination produces, and the hypothesis lets us divide by it.

Proof.

(a) By (F5), (F8) and (F7), $a(a^{-1}b) = (aa^{-1})b = 1 \cdot b = b$, so $x = a^{-1}b$ is a solution. If $ax = b$ and $ax' = b$, then $ax = ax'$ with $a \neq 0$, so $x = x'$ by cancellation (Theorem 2.1.1 (b)).

(b) Write $D := ad - bc$, so $D \neq 0$.

Uniqueness. Let (x, y) be a solution. Multiply the first equation by d, the second by b, and subtract. Multiplying out with (F9), reordering with (F5) and (F6), and using Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c) for the signs,

$$\begin{aligned} d(ax + by) - b(cx + dy) &= (ad)x + (bd)y - (bc)x - (bd)y \\ &= (ad - bc)x = Dx, \end{aligned}$$

while the right-hand sides give $de - bf = ed - bf$. So $Dx = ed - bf$. In the same way, $a(cx + dy) - c(ax + by) = (ad - bc)y = Dy$ and the right-hand sides give $af - ce$, so $Dy = af - ce$. By (a), applied twice with the non-zero coefficient D, $x = D^{-1}(ed - bf)$ and $y = D^{-1}(af - ce)$. These are the displayed values, since $D^{-1}u = uD^{-1} = u/D$ by (F6).

Existence. Put $x_0 := (ed - bf)D^{-1}$ and $y_0 := (af - ce)D^{-1}$. By (F5), (F6) and (F9), multiplying out and reordering with (F5) and (F6), and using Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c) for the signs,

$$\begin{aligned} ax_0 + by_0 &= \big(a(ed - bf) + b(af - ce)\big)D^{-1} \\ &= (ade - abf + abf - bce)D^{-1} \\ &= e(ad - bc)D^{-1} = eDD^{-1} = e, \end{aligned}$$

by (F8) and (F7) at the end. Likewise

$$\begin{aligned} cx_0 + dy_0 &= \big(c(ed - bf) + d(af - ce)\big)D^{-1} \\ &= (cde - bcf + adf - cde)D^{-1} \\ &= f(ad - bc)D^{-1} = f. \end{aligned}$$

So (x_0, y_0) is a solution, and by uniqueness it is the only one.

■

Both hypotheses have a job. Over $\mathbb{Z}$, part (a) fails: $2 \neq 0$, and $2x = 1$ has no integer solution, because the inverse 2^{-1} of (F8) is missing. And if $ad - bc = 0$, part (b) fails in every field: the system $x + y = 1$, $x + y = 0$ has no solution, since it would give $1 = 0$, while $x + y = 0$, $2x + 2y = 0$ has the solutions $(t, -t)$ for every $t \in F$, at least two of them ($t = 0$ and $t = 1$). The linear-algebra volumes treat n equations in n unknowns by the same elimination.

2.1.5 Examples

Three fields are already at hand. Each was proved a field in the sense of Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1, and the remark after Definition 2.1.1 translates.

Example. The Fields $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$

Explain why $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$, with their usual addition and multiplication, are fields.

Solution. For $\mathbb{Q}$, built as classes of pairs of integers, Number Systems: From \(\nN\) to \(\nR\), Theorem 3.1.3 proves that it is a field in the sense of Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1. For $\mathbb{R}$, built as classes of Cauchy sequences of rationals, Number Systems: From \(\nN\) to \(\nR\), Theorem 5.2.3 does the same; there $1 \neq 0$ is part of the field clause. For $\mathbb{C} := \mathbb{R} \times \mathbb{R}$, Analytic Preliminaries, Theorem 1.2.4 proves it, resting on the arithmetic laws Analytic Preliminaries, Proposition 1.1.2 and the inverse Analytic Preliminaries, Proposition 1.2.3. In each case the remark after Definition 2.1.1 turns "field in the sense of *Number Systems*" into "field in the sense of Definition 2.1.1", with the same zero, one, negatives and inverses.

These three are infinite: each contains an injective image of the infinite set $\mathbb{N}$ (Sets and Functions, Example (Hilbert's Hotel)), through Number Systems: From \(\nN\) to \(\nR\), Proposition 2.2.3 (a), Number Systems: From \(\nN\) to \(\nR\), Proposition 3.1.4, Number Systems: From \(\nN\) to \(\nR\), Theorem 5.4.1 (b) and Analytic Preliminaries, Proposition 1.1.1. A finite set has only finite subsets (Sets and Functions, Proposition 4.1.6). The next section produces finite fields, and Section 2.3 produces fields strictly between $\mathbb{Q}$ and $\mathbb{R}$.

Changing an operation on a familiar set can give a new field; deciding means checking every clause.

Example. A New Multiplication on $\mathbb{Q}$

Show that $\mathbb{Q}$, with the usual addition and the new multiplication $a \odot b := -(ab)$, is a field, and find its one.

Solution. Addition is that of $\mathbb{Q}$, so (F1)–(F4) hold with the usual 0 and negatives. The new product is a function $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$. Each identity below is computed in the field $\mathbb{Q}$ (Example (The Fields $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$)), the signs by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a) and (c). For all $a, b, c \in \mathbb{Q}$:

- (F5): $(a \odot b) \odot c = -((-(ab))c) = abc = -(a(-(bc))) = a \odot (b \odot c)$.
- (F6): $-(ab) = -(ba)$.
- (F9): $a \odot (b + c) = -(ab + ac) = -(ab) + (-(ac)) = a \odot b + a \odot c$, since $-(u + v) = (-1)(u + v) = -u + (-v)$ by Theorem 2.1.1 (e).
- (F7): the one is -1, since $a \odot (-1) = -(a(-1)) = -(-a) = a$, and $-1 \neq 0$ (otherwise $1 = -(-1) = -0 = 0$, where $-0 = 0$ by Theorem 2.1.1 (c), since $0 + 0 = 0$).
- (F8): a non-zero a has the inverse a^{-1}, the usual one, since $a \odot a^{-1} = -(aa^{-1}) = -1$, the one of this field.

The one of a field need not be the number 1.

2.1.6 Non-examples

Each system below falls short of the slogan somewhere. For each, find the precise clause that breaks; a vague "it is not like $\mathbb{Q}$" teaches nothing.

Example. Four Systems That Are Not Fields

For each of the following, with its usual operations, name the clauses of Definition 2.1.1 that fail, with a witness: (i) $\mathbb{N}$; (ii) $\mathbb{Z}$; (iii) the zero ring $\{0\}$, with $0 + 0 := 0$ and $0 \cdot 0 := 0$; (iv) $\mathbb{Z}/6\mathbb{Z}$.

Solution.

(i) *Number Systems* checked the minimal change from $\mathbb{N}$ to $\mathbb{Z}$: $\mathbb{N}$ satisfies (F1)–(F3), (F5)–(F7) and (F9), and (F4) fails at $a = 1$, since $1 + x \neq 0$ for every $x \in \mathbb{N}$ (the non-example after Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1). The condition $1 \neq 0$ holds, since it holds in $\mathbb{Z}$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 2.3.1), and $\mathbb{N} \subseteq \mathbb{Z}$ with ι injective (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.2.3 (a)). Clause (F8) fails too, at $a = 2$: a natural number x with $2x = 1$ would be an integer with $2x = 1$ (ι preserves products, Number Systems: From \(\nN\) to \(\nR\), Proposition 2.2.3 (a)), and there is none (Number Systems: From \(\nN\) to \(\nR\), Corollary 2.5.3 (a)).

(ii) $\mathbb{Z}$ is a commutative ring with $1 \neq 0$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 2.3.1), so (F1)–(F7) and (F9) hold. Only (F8) fails, at $a = 2$, by the same parity argument.

(iii) Every clause other than (F7) and (F8) is an equation between elements of $\{0\}$, and each such equation reads $0 = 0$; (F8) holds because there is no non-zero element to invert. The only candidate for the element 1 of (F7) is 0, and (F7) demands $1 \neq 0$. So exactly the condition $1 \neq 0$ inside (F7) fails. This is the reason the condition is there: without it, a one-element "field" would make every statement of the form "a field has a non-zero element" false.

(iv) $\mathbb{Z}/6\mathbb{Z}$ is a commutative ring with $[1] \neq [0]$ (Theorem 1.3.3 (a), (c)), so (F1)–(F7) and (F9) hold. But $[2][3] = [6] = [0]$ (Example (Multiplication modulo six and modulo

five)), while $[2] \neq [0]$ and $[3] \neq [0]$, since 6 divides neither 2 nor 3. If $[2]$ had an inverse u, then $[3] = (u[2])[3] = u([2][3]) = u[0] = [0]$, by (F7), (F8), (F6), (F5) and Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (b), which is false. So (F8) fails at $[2]$. Had $\mathbb{Z}/6\mathbb{Z}$ been a field, Theorem 2.1.1 (f) would have forbidden the product $[2][3] = [0]$ outright.

The first linear-algebra volume meets one more kind of non-example, the square matrices, where (F6) fails: two matrices can give different products in the two orders.

2.1.7 Why these axioms

Drop a clause and something we use breaks.

- **Without (F8)**, elimination halts at the first coefficient that cannot be inverted, as $2x = 1$ over $\mathbb{Z}$ showed.
- **Without (F6)**, numbers would not commute with each other. Linear algebra swaps scalars constantly (in the first linear-algebra volume, when a vector is multiplied by two numbers in turn), so commutativity is kept.
- **Without** $1 \neq 0$, the zero ring of Example (Four Systems That Are Not Fields) (iii) would count as a field.

The word "field" is a historical label; nothing in it is meant literally.

Warning. Order is not part of a field. Clauses (F1)–(F9) say nothing about $<$, so "squares are ≥ 0" has no meaning in a general field, and read in $\mathbb{C}$ it is false, since $i^2 = -1$ (*Analytic Preliminaries*, Chapter 1). Likewise "$1 + 1 \neq 0$" is not a clause, and it fails in the field with two elements of the next section. In a general field, never use an inequality.

2.1.8 Exercises

2.1.8.1 A. Check your understanding

Exercise. A1

(a) State the definition of a field.

(b) Which clauses of Definition 2.1.1 does each of $\mathbb{N}$, $\mathbb{Z}$ and the zero ring $\{0\}$ fail?

(c) True or false: in every field, $ab = ac$ implies $b = c$. Justify your answer.

(d) Which parts of Theorem 2.1.1 fail in the commutative ring $\mathbb{Z}/6\mathbb{Z}$? Give a witness for each.

(e) Use Proposition 2.1.3 to solve $3x + 2y = 1$, $x - y = 2$ in $\mathbb{Q}$.

Solution.

(a) A field is a set F with two functions $+, \cdot : F \times F \to F$ satisfying (F1)–(F9) of Definition 2.1.1, where (F7) includes $1 \neq 0$ and (F8) asks for inverses of non-zero elements only.

(b) $\mathbb{N}$: (F4), at $a = 1$, and (F8), at $a = 2$. $\mathbb{Z}$: only (F8), at $a = 2$. $\{0\}$: only the condition

$1 \neq 0$ in (F7). (See Example (Four Systems That Are Not Fields).)

(c) False. With $a = 0$, $b = 1$, $c = 2$ in $\mathbb{Q}$: $0 \cdot 1 = 0 = 0 \cdot 2$, but $1 \neq 2$. Cancellation needs $a \neq 0$ (Theorem 2.1.1 (b)).

(d) Multiplicative cancellation in (b) fails: $[2][1] = [2]$ and $[2][4] = [8] = [2]$, but $[1] \neq [4]$, since 6 does not divide $4 - 1 = 3$. Part (f) fails: $[2][3] = [0]$ with $[2], [3] \neq [0]$. The first clause of (g) fails for the same pair: $[2] \neq [0]$, $[3] \neq [0]$, but $[2][3] = [0]$. The uniqueness of inverses in (c) is not violated, but for $[2]$ it says nothing, since $[2]$ has no inverse (Example (Four Systems That Are Not Fields) (iv)). Parts (a), (d) and (e) hold in every commutative ring (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a)–(c), with (F6), (F7)), and so does additive cancellation, whose proof in (b) uses only (F1)–(F4).

(e) Here $a = 3$, $b = 2$, $c = 1$, $d = -1$, $e = 1$, $f = 2$, and $ad - bc = -3 - 2 = -5 \neq 0$. So

$$x = \frac{1 \cdot (-1) - 2 \cdot 2}{-5} = \frac{-5}{-5} = 1, \qquad y = \frac{3 \cdot 2 - 1 \cdot 1}{-5} = \frac{5}{-5} = -1.$$

Check: $3 \cdot 1 + 2 \cdot (-1) = 1$ and $1 - (-1) = 2$.

2.1.8.2 B. Practice

Exercise. B1

Let F be a field and $a, b, x \in F$. Prove:

(a) $-(a - b) = b - a$;

(b) if $a \neq 0$, then $-a \neq 0$ and $(-a)^{-1} = -(a^{-1})$;

(c) $x^2 = x$ if and only if $x = 0$ or $x = 1$;

(d) $x^2 + x = 0$ if and only if $x = 0$ or $x = -1$.

Solution.

(a) By (F1) and (F2), $(a - b) + (b - a) = (a + (-a)) + (b + (-b)) = 0 + 0 = 0$, using (F4) and (F3). So $b - a$ satisfies the defining equation of $-(a - b)$, and $-(a - b) = b - a$ by the uniqueness of negatives (Theorem 2.1.1 (c)).

(b) If $-a = 0$, then $a = -(-a) = -0 = 0$, by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a) and $0 + 0 = 0$; so $-a \neq 0$. By Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c) and (F8), $(-a)(-(a^{-1})) = aa^{-1} = 1$. So $-(a^{-1})$ satisfies the defining equation of $(-a)^{-1}$, and the two are equal by Theorem 2.1.1 (c).

(c) By (F7), (F9) and Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c), $x(x - 1) = xx + x(-1) = x^2 - x$. So $x^2 = x$ if and only if $x(x - 1) = 0$, adding $-x$ or x to both sides. By Theorem 2.1.1 (f), $x(x - 1) = 0$ gives $x = 0$ or $x - 1 = 0$, that is, $x = 1$. Conversely, $0^2 = 0$ by Theorem 2.1.1 (d), and $1^2 = 1$ by (F7).

(d) By (F7) and (F9), $x(x + 1) = x^2 + x$. So $x^2 + x = 0$ means $x(x + 1) = 0$, which by Theorem 2.1.1 (f) gives $x = 0$ or $x + 1 = 0$, and the latter means $x = -1$ by the uniqueness of negatives (Theorem 2.1.1 (c)). Conversely, $0^2 + 0 = 0$ by Theorem 2.1.1 (d) and (F3), and $(-1)^2 + (-1) = 1 + (-1) = 0$, since $(-1)(-1) = 1 \cdot 1 = 1$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c).

Exercise. B2

Let F be a field and $a, b, c, d \in F$ with $b \neq 0$ and $d \neq 0$. Prove that $a/b = c/d$ if and only if $ad = bc$.

Solution. ($\Rightarrow$) Multiply both sides of $ab^{-1} = cd^{-1}$ by bd. Reordering with (F5) and (F6), and using (F8) and (F7), the left side becomes $a(b^{-1}b)d = ad$ and the right side $c(d^{-1}d)b = cb = bc$. So $ad = bc$.

($\Leftarrow$) By Theorem 2.1.1 (g), $bd \neq 0$ and $(bd)^{-1} = b^{-1}d^{-1}$. Multiply both sides of $ad = bc$ by $b^{-1}d^{-1}$: reordering as before, the left side becomes $ab^{-1}(dd^{-1}) = ab^{-1}$ and the right side $(bb^{-1})cd^{-1} = cd^{-1}$. So $a/b = c/d$.

Exercise. B3: Which Are Fields?

Determine which of the following are fields. Justify your answer.

(a) $\mathbb{Q}$, with the usual addition and the new multiplication $a \odot b := 2ab$.

(b) $\mathbb{Q}$, with the usual addition and the new multiplication $a \odot b := ab + 1$.

(c) The set $\{0, 1\} \subseteq \mathbb{Q}$, with the addition and multiplication of $\mathbb{Q}$.

Solution.

(a) **A field.** Addition is that of $\mathbb{Q}$, so (F1)–(F4) hold with the usual 0 and negatives. The new product is a function $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$. For all $a, b, c \in \mathbb{Q}$: (F5) $(a \odot b) \odot c = 2(2ab)c = 4abc = 2a(2bc) = a \odot (b \odot c)$; (F6) $2ab = 2ba$; (F9) $a \odot (b + c) = 2a(b + c) = 2ab + 2ac = a \odot b + a \odot c$. For (F7) the one is $\frac{1}{2}$: $a \odot \frac{1}{2} = 2a \cdot \frac{1}{2} = a$, and $\frac{1}{2} \neq 0$. For (F8), a non-zero a has the inverse $\frac{1}{4a}$, since $a \odot \frac{1}{4a} = \frac{2a}{4a} = \frac{1}{2}$, the one of this field. Each identity is computed in the field $\mathbb{Q}$ (Example (The Fields $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$)). The one of a field need not be the number 1; Exercise B3 of Section 2.4 shows that this field is $\mathbb{Q}$ in disguise.

(b) **Not a field:** (F9) fails. For all a, b, c, $a \odot (b + c) = ab + ac + 1$, while $a \odot b + a \odot c = ab + ac + 2$, and $1 \neq 2$. For instance $a = b = c = 0$ gives $1 \neq 2$.

(c) **Not a field:** addition is not a function $\{0, 1\} \times \{0, 1\} \to \{0, 1\}$, since $1 + 1 = 2 \notin \{0, 1\}$. (The next section puts a different addition on a two-element set and obtains a field.)

2.1.8.3 C. Going deeper

Exercise. C1: Exit check: rules from the axioms

Let F be a field and $a, b, x, y \in F$. Prove the following from (F1)–(F9), using of Theorem 2.1.1 only part (c) (uniqueness of negatives and inverses) and part (e) ($(-1)a = -a$), and citing no result of *Number Systems*.

(a) $0 \cdot a = 0$.

(b) If $ab = 0$, then $a = 0$ or $b = 0$.

(c) $x^2 = y^2$ if and only if $x = y$ or $x = -y$.

Solution.

(a) By (F3), $0+0=0$, so by (F9), $a\cdot 0 = a(0+0) = a\cdot 0 + a\cdot 0$. Add $-(a\cdot 0)$, which exists by (F4), to both sides. On the left this gives 0. On the right, by (F2), (F4) and (F3), $(a\cdot 0 + a\cdot 0) + (-(a\cdot 0)) = a\cdot 0 + (a\cdot 0 + (-(a\cdot 0))) = a\cdot 0 + 0 = a\cdot 0$. So $0 = a\cdot 0$, and $0\cdot a = a\cdot 0 = 0$ by (F6).

(b) Suppose $ab = 0$. If $a = 0$, we are done. Otherwise a^{-1} exists by (F8), and by (F7), (F8), (F5) and (F6),

$$b = b\cdot 1 = b(aa^{-1}) = (ba)a^{-1} = (ab)a^{-1} = 0\cdot a^{-1} = 0,$$

the last step by (a).

(c) *Two sign rules.* For $u, v \in F$, part (e), (F5) and (F6) give $(-u)v = ((-1)u)v = (-1)(uv) = -(uv)$. Hence $(-1)(-1) = -(1\cdot(-1)) = -(-1) = 1$, where $1\cdot(-1) = -1$ by (F6), (F7), and $-(-1) = 1$ by part (c), since $(-1)+1 = 0$ by (F1), (F4). So $(-y)(-y) = ((-1)y)((-1)y) = ((-1)(-1))(yy) = yy$, by (F5) and (F6).

The factorization. By (F9), (F6) and the first sign rule,

$$\begin{aligned}(x+(-y))(x+y) &= xx + xy + (-y)x + (-y)y \\ &= xx + xy + (-(xy)) + (-(yy)),\end{aligned}$$

using $yx = xy$ by (F6). By (F2), (F4) and (F3) the middle terms cancel, so $(x-y)(x+y) = x^2 - y^2$.

($\Rightarrow$) If $x^2 = y^2$, then $x^2 - y^2 = 0$ by (F4), so $(x-y)(x+y) = 0$, and (b) gives $x - y = 0$ or $x + y = 0$. From $x - y = 0$, adding y and using (F2), (F1), (F4), (F3) gives $x = y$. From $x + y = 0$, x satisfies the defining equation of $-y$ (with (F1)), so $x = -y$ by part (c).

($\Leftarrow$) If $x = y$ there is nothing to prove; if $x = -y$, then $x^2 = (-y)(-y) = y^2$ by the sign rules.

Marking scheme. (10 points)

- 2: for (a), writes $0 = 0+0$ and multiplies out $a(0+0)$ by (F9).
- 2: cancels $a\cdot 0$ by adding its negative, naming (F2), (F4) and (F3).
- 1: obtains $0\cdot a$ from $a\cdot 0$ by (F6).
- 2: for (b), treats the case $a = 0$ separately and spends $a \neq 0$ on (F8).
- 1: writes the chain $b = b\cdot 1 = \cdots = 0$ with each clause named.
- 1: proves $(x-y)(x+y) = x^2 - y^2$, justifying the cancellation of xy against $-(yx)$, and concludes by (b).
- 1: proves the converse, deriving $(-y)(-y) = yy$ from part (e), (F5) and (F6).

Fatal errors (0 for the step): dividing by a without $a \neq 0$; "$0\cdot a = 0$ is obvious"; citing (b) of the exercise inside the proof of (a). A proof of (c) that factors as $(x+y)(x-y)$ or that cites (b) as "no zero divisors" earns the same points.

Exercise. C2: Find the error

Here is a claim with a proof.

Claim. In a field F, if $x \in F$ and $x^2 = x$, then $x = 0$ or $x = 1$.

Proof. (1) Let $x^2 = x$. (2) Dividing both sides by x gives $x = 1$. (3) So $x = 0$ or $x = 1$.

Is the claim true? Does the proof prove it? If not, name the line that fails, the error, and give a counterexample to the step.

Solution. The **claim is true**: it is Exercise (B1) (c).

The **proof is wrong**, at line (2). Dividing by x means multiplying by x^{-1}, which (F8) supplies only when $x \neq 0$. No line provides $x \neq 0$, so the step cites an unavailable "fact", cancellation of x without $x \neq 0$: entry (N9) of Logic and Proof, Example (What is not allowed). Line (3) is a correct weakening of (2) by $\vee$I; the proof fails only at (2).

A counterexample to the step: $x = 0$ satisfies $x^2 = x$, and "$x = 1$" is false for it, in every field. The step's conclusion fails, although the claim's does not.

A repair: $x^2 = x$ gives $x(x - 1) = 0$, and Theorem 2.1.1 (f) gives $x = 0$ or $x - 1 = 0$, which treats both cases at once.